

PHARMACOLOGY: PHARMACODYNAMICS

Practice Quiz

➤ Pharmacology bridges physiology, pathophysiology, and clinical practice.

Science of Successful Learning:

Smart Study Tips

Set specific and *achievable* goals that will focus and track your progress.

- Practice retrieval: Test yourself regularly.
- Space out your study: Studying material repeatedly over time helps your brain consolidate information and is more effective than cramming.
- Mix up the topics: Switching between subjects and types of problems strengthens memory retrieval and promotes better conceptual understanding.
- Elaborate: Ask yourself **why** and **how** questions to add meaning to your learning and create richer mental connections.
- Dual coding: Draw and label figures, diagrams, flowcharts, etc.

Get enough sleep and drink enough water.

Goals of Practice Quizzes:

1. To help students identify what they know and what they understand by applying pharmacology knowledge to clinical case vignettes.
2. To help students get used to thinking through the “Board-style” case vignettes and questions. NYITCOM exams, COMLEX, and USMLE test students’ medical knowledge using these types of questions.

Note: Some items may have more than one answer.

My recommendation is to home in on the information you need to answer the question, which, at this time, is related to pharmacodynamics. *Don’t let yourself get distracted by the background information that is not relevant for answering the question.*

CHALLENGE: Cover the options and try to answer the questions without looking at the options.

YOU CAN EASILY FIND THE ANSWERS.

“Search” my Notes Handout (“Notes”). All answers can be found there. (You do not need to read the whole document if you do not wish to.)

If you prefer reading the textbook, links are provided on the PowerPoint and in the CPG.

Putting in the effort to find the answers cements learning much better than rote memorization.

Answer Key: Pharmacology concepts are explained in a physiologic and clinical contexts.

THE ANSWER KEY with explanations will be POSTED Friday November 21.

First, the basics.

1. The pharmacokinetics and pharmacodynamics properties of a drug influence the dose-response relationship of that drug. What feature of rational drug dosing links the two properties?

- A. Absorption
- B. Concentration*
- C. Intrinsic activity
- D. Potency

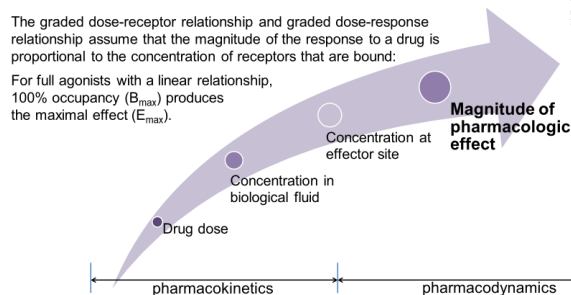
Hint: Notes page 2

Rationale: B. concentration

Concentration provides the link between pharmacokinetics and pharmacodynamics and is the focus of the **target concentration approach to rational dosing**:

Dose → plasma concentration → concentration at effector site → magnitude of effect

- The dose determines the concentration of a drug in plasma, which determines the concentration of the drug at the effector site, which determines the magnitude of the effect.
- Furthermore, the dose-plasma concentration determines the fraction of receptors in the total receptor pool that will be occupied by the drug:
 - with a lower dose → a smaller fraction of receptors is occupied;
 - with a higher dose → a higher fraction of receptors is occupied



❖ The graded dose-receptor relationship and dose-response relationship assume the magnitude of the response is proportional to the concentration of receptors bound.

Incorrect options:

- A. Absorption, distribution, metabolism, and excretion (ADME processes) determine the drug **concentration** in plasma and **rate of drug elimination**, which in turn affects the concentration of drug in plasma – as the drug is eliminated (metabolism/excretion), the plasma concentration is reduced, if dosing does not continue.
- C. Intrinsic activity is defined as a drug's ability to activate a receptor and its signaling cascade. Intrinsic activity is determined by the physicochemical properties of the drug. Antagonists have zero intrinsic activity yet can have a small to large effect depending on the concentration of the antagonist.
- D. Potency, the amount (dose) of a drug needed to produce the desired effect, is related to its affinity for the biological molecule. Drugs with high affinity are more potent than drugs with lower affinity. Yet, it is the *concentration* of drug molecules at the site of action that will determine the *magnitude* of effect.

2. Finish this sentence: A drug with a low K_d would have high_____

- A. Affinity*
- B. Efficacy
- C. Functional response
- D. Potency
- E. Specificity

Hint: Notes page 5

Rationale: A. Affinity

- K_d is the equilibrium dissociation constant: Half of the receptors are unoccupied (free) and half are bound by drug (ligand).
- Thus, K_d represents the drug concentration required to saturate 50% of the receptors:
 $D + R \rightleftharpoons DR \rightarrow \text{Response}$
 $K_d = ([D]/[R]) / [DR] = k_{\text{off}}/k_{\text{on}}$
- The affinity constant is the reciprocal of K_d : $K_a = 1/K_d$. K_a is inversely proportional to K_d .
- The smaller the K_d , the higher the affinity. That is, smaller K_d drugs have a relatively slow rate of dissociation (k_{off}) from the receptor compared to drugs with lower affinity.
- Conversely, the higher the K_d the lower the affinity – it will dissociate more rapidly compared to drugs with higher affinity.

Incorrect options:

- **B. Efficacy** is the quantitative ability of the drug to produce a biologic effect – the greatest effect (E_{max}) an agonist can produce if the dose is taken to the highest tolerated level.
Dose → plasma concentration → concentration at effector site → magnitude of effect
- **C. Functional response** to a drug is largely determined by the dose and plasma concentration of the drug – AND on the receptor-drug interaction properties – affinity and intrinsic activity
- **D. Potency** is the concentration of drug (EC_{50}) required to produce that drug's maximal effect.
Potency and affinity are related by different. A drug with high affinity binds tightly to the receptor and is more likely to stay attached longer (slower dissociation rate). Therefore, a lower dose of the drug is needed to produce a given effect – the drug is more potent – than a drug with lower affinity.
- **E. Specificity** is the degree to which a particular drug gains access to and binds a *select receptor type* and induces a conformational change that activates or inhibits a biochemical or physiologic effect. A drug with low specificity binds more than one receptor type.

3. Fill in the blank: EC50 is useful for measuring the relative _____ of a two or more drugs.

- A. Potency*
- B. Affinity
- C. Efficacy
- D. Functional response
- E. Specificity

Hint: Page 7 of Notes Handout

Rationale: A. potency

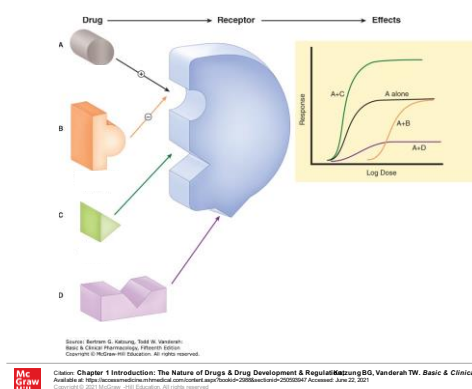
- Potency refers to the amount (the dose, the concentration) of drug required to produce that drug's maximal effect compared to other drugs producing the same effect.
- EC50 quantifies drug activity. EC50 is defined as the concentration of drug that produces 50% of that drug's maximal response

When two (or more) drugs produce the same therapeutic action, the drug that provokes the response at a lower concentration (lower EC50) is more potent than the drug that elicits the same response at a higher concentration (higher EC50).

❖ Pharmacology principle: Potency is a comparative expression in pharmacology terminology. (In everyday language, potent is often used alone, meaning the drug is effective at low doses.)

4. Looking at the figure on the right, which one of the following options graphically describes a competitive antagonist?

- A. A alone
- B. A+B*
- C. A+C
- D. A+D



Hint:

The models of the interactions of two drugs at a given receptor is depicted on Part 2, Slide 15 and in the Notes on pages 12 and 13.

The figure above right shows a different way of illustrating the same concept. Try explaining to yourself and your study mates what A, B, C, and D represent and then the effect of each as represented in the graph.

If you want to check yourself, the illustration is explained in the Katzung textbook, Chapter 1, Figure 1-1: <https://accessmedicine-mhmedical-com.nyit.idm.oclc.org/content.aspx?bookid=3461§ionid=285588535>

Rationale: B. A+B

The effect of a competitive antagonist (B on the figure) is represented by A+B on the graph. The competitive antagonist reversibly binds to the active (agonist) site of a receptor (or the catalytic site of an enzyme) and prevents (inhibits) the natural ligand or drug agonist from gaining access to the site.

High concentrations of agonist can overcome the antagonistic actions of the competitive antagonist → The EC₅₀ is shifted to the right ⇒ The potency of the agonist is reduced.
Agonist = natural ligand or drug agonist

The other options:

- A. A alone. The agonist and the dose-response curve of the agonist.
- C. A+C: Drug C is a positive allosteric modulator (also called allosteric activator).
Drug C binds to an allosteric site (near, but not in, the agonist site), changes the conformation of the agonist site, which enhances the affinity.
In the figure, the EC₅₀ is slightly shifted to left compared to the agonist (A) alone, indicating increased potency, and the E_{max} is higher, indicating a greater response (the response is amplified).
Notice that the dose-response curve is shifted to the right (lower EC₅₀) and the plateau is higher than that of A, the agonist alone, indicating a larger response = greater efficacy, E_{max}.
- D. A+D: Drug D is an allosteric inhibitor. Binding of the inhibitor to an allosteric site reduces the maximal activity of the receptor.

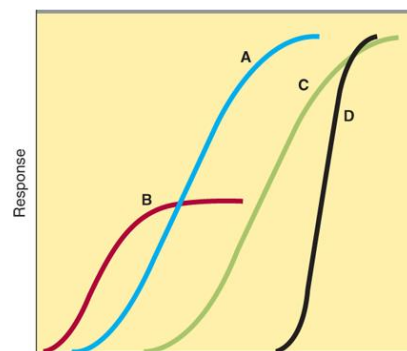
5. An investigational opioid analgesic agent, Drug B, undergoes preclinical testing to determine its potency and efficacy compared to the currently available analgesic drugs, Drugs A, C, D, in accordance with US Food and Drug Administration (FDA) regulations. They all have the same therapeutic mechanism of action. The graded dose-response curves are shown in the graph.

Which drug is the most potent?

Short answer.

Hint: Apply your knowledge of relative EC₅₀ values to answer this question.

If you want to check yourself, see Notes p8.



Source: Bertram G. Katzung, Todd W. Vanderah:
Basic & Clinical Pharmacology, Fifteenth Edition
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Rationale: B is the correct answer.

Potency is a measure of the amount of drug (the concentration) necessary to produce an effect.

Legend on Katzung, Figure 2-15 accompanying the question:

Visualize, or trace with your pencil, the approximate EC₅₀s of the four drugs.

- Drugs A and B are said to be more potent than drugs C and D because of the relative positions of their dose-response curves along the dose axis (x axis) of Figure 2–15.
- Potency refers to the concentration (EC₅₀) (sometimes referred to as the dose, ED₅₀) of a drug required to produce 50% of that drug's maximal effect.
- Thus, the pharmacologic potency of drug A in Figure 2–15 is less than that of drug B because the EC₅₀ of A is greater than the EC₅₀ of B.
 - ❖ Please note: Although drug B is a partial agonist (it does not achieve the full effect), it is the most potent of the four drugs, whereas drugs A, C, and D produce larger effects (maximal efficacy, E_{max}) than any dose of drug B.
 - ❖ A drug with a steep dose-response curve (drug D) indicates that a large effect is produced with a relatively small increase in dose. This sharp increase in effect may result in serious clinical consequences due to drug toxicity, if the upper portion of the curve represents an undesirable extent of response – for example, respiratory depression caused by an opioid analgesic.

6. Benzodiazepines are positive allosteric modulators at GABA-A receptors. GABA-A receptors are ion channels with 5 subunits and are activated by the neurotransmitter, GABA. GABA binds to its two active sites at the interface between the α and β subunits, allowing negatively charged Cl⁻ ions to move down its electrochemical gradient into the cell, resulting in hyperpolarization – which leads to inhibition of neuronal firing.

i) What type of receptor is the target of the benzodiazepines?

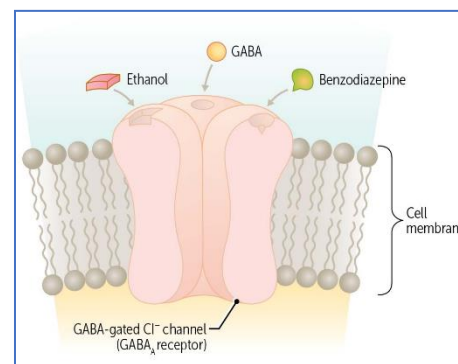
- A. Ligand-gated ion channel*
- B. Nuclear receptor
- C. Receptor tyrosine kinase
- D. Solute carrier
- E. Voltage-gated ion channel

ii) What is the effect of benzodiazepines (simple answer)?

Note: Please try to deduce the answer from the description even though the this receptor is not specifically mentioned in the lecture or Notes.

Rationale: A. ligand-gated ion channel / Enhanced inhibitory effect

i) Gamma-amino butyric acid, GABA, is the ligand.



ScholarRx

The ligand GABA is a major inhibitory neurotransmitter in the central nervous system. When activated, GABA-A receptors allow chloride ions (primarily) to flow into the neuron, causing a hyperpolarization and thus inhibiting the neuron from propagating signals.

ii) Benzodiazepines enhance (or potentiate) the effect of GABA.

Benzodiazepines bind an allosteric site on GABA-A channels, enhancing the receptor's response to GABA by increasing the frequency of channel opening. This action potentiates the inhibitory effect of GABAergic transmission.

Physiology:

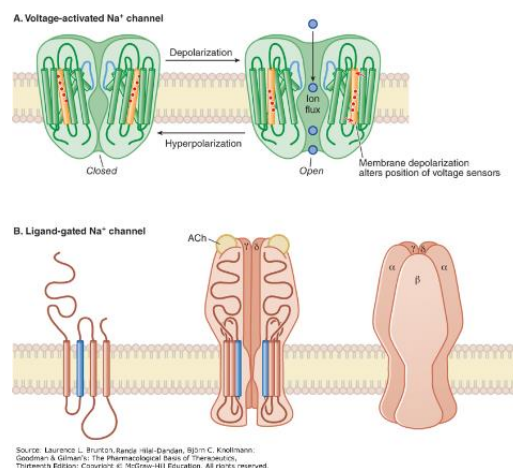
Ligand-gated ion channels are transmembrane ion channels activated by the binding of a molecule – ligand – opens and allows ions to pass through the cell membrane.

The pores of GABA-A channels are structured to allow negatively charged chloride to pass through, which hyperpolarizes the cell, making the cell less likely to fire an action potential, which exerts an inhibitory effect.

Ligand-gated cation channels, such as nicotinic acetylcholine receptor, allow sodium, potassium, or sometimes calcium ions through and depolarize the cell. If the critical threshold is reached, an action potential is initiated.

Incorrect options:

- B. Nuclear receptors comprise a class of intracellular proteins that regulate gene expression in response to small lipophilic molecules like steroid hormones, thyroid hormones, and other signaling molecules.
- C. Receptor tyrosine kinases are high-affinity cell surface receptors that bind to various growth factors, cytokines, and hormones. Receptor activation causes dimerization, autophosphorylation, and activation of intracellular signaling.
- D. Solute carriers are membrane transporters of the SLC superfamily that transport mainly hydrophilic substances.
- E. Voltage-gated ion channels (figure right) are transmembrane proteins activated by changes in the electrical membrane potential.



Scholar Rx

Note: If you are interested, voltage-gated ion channels are described ScholarRx Neurology and Special Senses > Brick: Action Potentials <https://exchange.scholarrx.com/brick/action-potentials>

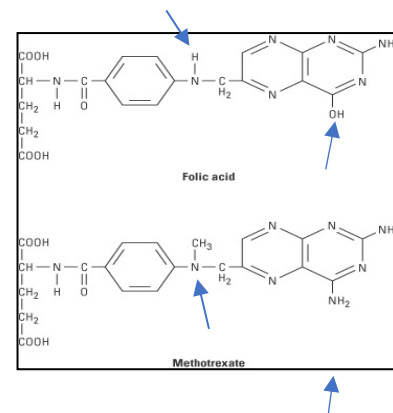
7. A 7-year-old child's treatment for acute lymphoblastic leukemia includes methotrexate (lower structure on the figure), which blocks the activation of folic acid (upper molecule) in the cancer cells. This action leads to cancer cell apoptosis. Methotrexate is a structural analog of folic acid.

The arrows are pointing to the differences in the molecules.

This description is an example of what pharmacologic concept?

Short answer

Hint: Think about the pharmacologic concept exemplified by the figure.



Rationale: Structure-activity relationship

In this case, slight alterations in the structure of folic acid (a substrate of the enzyme dihydrofolate reductase DHFR) creates the antagonist methotrexate. Methotrexate competitively inhibits folic acid from binding to the active site of DHFR, ultimately preventing the synthesis of DNA.

➤ Review: Pharmacokinetics – Pharmacodynamics bridge:

The physicochemical characteristics of a drug determine its ability to cross physiologic barriers, move through the body (pharmacokinetic property of distribution), and initiate a cellular effect (pharmacodynamics).

Structural features, such as size, shape, chirality, electrical charge, and lipid solubility determine:

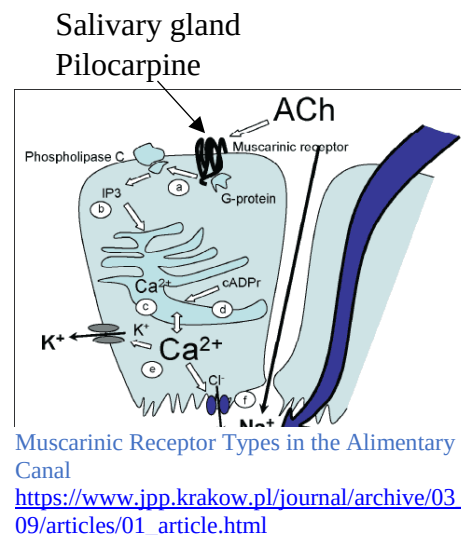
- Pharmacokinetic properties: A drug's ability to be **absorbed**, to **distribute** to the site of action, be **metabolized** and be **excreted**
- Pharmacodynamic properties: **Receptor specificity**, **receptor affinity**, and **intrinsic activity**

8. A 57-year-old woman with primary Sjogren's syndrome complains that the saliva substitute the nurse practitioner suggested isn't helping with her dry mouth. She agrees to take oral pilocarpine, which induces saliva secretion by binding to the muscarinic M3 receptors on the surface of secretory acini cells of salivary glands. The drug's effect is equivalent to that produced by the natural ligand of M3 receptors, acetylcholine (ACh).

i) What is the mechanism of action of the prescribed drug?

ii) Draw a graphic representation and label it.

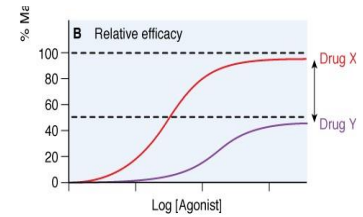
Definition: Sjogren's syndrome is a slowly progressing, chronic autoimmune disease affecting exocrine glands.



Rationale: Activation of the receptor / Full agonist action

Thinking it through:

- Pilocarpine binds to M3 receptors. Thus, the drug has affinity.
- Pilocarpine activates the receptor (induces saliva secretion). Thus, the drug has efficacy – the drug is an agonist.
- Pilocarpine produces an effect that is equivalent to the natural ligand's effect. Thus, the drug is a full agonist – Drug X in the dose-response curve (right).



Source: Brunton LL, Chabner BA, Knollmann BC: Goodman & Gilman's The Pharmacological Basis of Therapeutics, 12th Edition: www.accessmedicine.com Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Goodman & Gilman 2018 Figure 3-7

In the classical receptor occupation model, the receptors in a receptor pool are quiet (quiescent) unless activated by an agonist.

9. A drug development company ran a preclinical trial to evaluate an experimental antiarrhythmic medicine among male and female mouse models of atrial fibrillation. Endpoints were improved functional status (efficacy) and onset of arrhythmia (toxicity). The percentage of individuals that showed the defined effect at each dose level was recorded.

(This is a fictional scenario to my knowledge.)

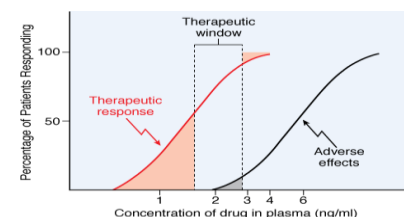
Which of the following curves would BEST describe the findings in this study?

- A. Drug concentration-time plot
- B. Receptor binding-concentration curve
- C. Graded dose-response curve
- D. Lineweaver-Burk plot
- E. Quantal dose-response curve*

Rationale: E. Quantal dose-response curve (also called quantal dose-effect curve)

Thinking it through:

- The vignette describes a defined therapeutic effect and a defined toxic effect of a drug.
- The outcome is binary – an all-or-none response (quantal relationship).
- The percentage of individuals that showed the defined effect at each dose level was recorded.
- The data was plotted as a cumulative dose-response curve.



Source: Brunton LL, Chabner BA, Knollmann BC: Goodman & Gilman's The Pharmacological Basis of Therapeutics, 12th Edition: www.accessmedicine.com Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Goodman & Gilman Figure 3-10

Incorrect options:

- A. Drug concentration v. time curves plot the rate of elimination of a dose of drug in one individual over time.

They are used for determining pharmacokinetic parameters (ADME), optimizing drug dosing for the individual patient, and therapeutic drug monitoring, among others.

- B. Drug-receptor binding v. concentration curve are useful for demonstrating the receptor occupation with increasing doses. The parameters are:
 - K_d – drug concentration at which drug-receptor bound and unbound drug are in equilibrium
 - B_{max} – maximal receptor occupation
- C. Graded dose-response curves plot responses to increasing doses to understand how a drug's effect in an individual changes with various doses.

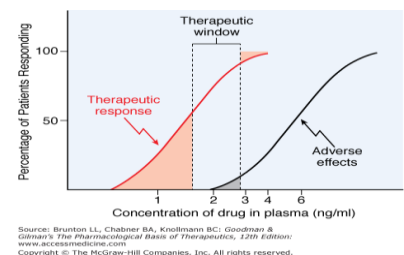
They are used to compare the magnitude of therapeutic effect (efficacy, E_{max}) and potency (using EC_{50}) of different drugs.
- D. Lineweaver-Burk plot is a double-reciprocal plot used to estimate K_m and V_{max} and to analyze enzyme inhibition.

10. What is the clinical application of plotting the curves described in the above vignette?

- A. Determining the maximal effect of the drug
- B. Determining the elimination rate of the drug
- C. Determining the relative potency of the drug
- D. Determining the effective dose range of the drug*
- E. Determining the safe dose range of the drug*

Rationale: D and E. effective and safe dose range

Quantal dose-response curves may be used to determine the **efficacy and safety** of a drug and to establish dosing guidelines and labeling for prescription drugs.



- ❖ **Pharmacology Pearl:** Quantal dose-response curves are useful for determining the range of doses over which the drug is effective for most of the population and with acceptable toxicity.
- The cumulative dose-response curves show the range of doses that produce a therapeutic effect and the range of doses that produce a toxic effect (or lethal effect).
- The cumulative dose-response curves may be used to describe several parameters for safe use of drugs, such as the ED_{50} , LD_{50} , TD_{50} , and margin of safety.
- Therapeutic window is defined as range between the minimum effective and the minimum toxic dose.

- **Practice point:** The therapeutic dose can be achieved in the lower portion of the therapeutic window for the majority of drugs and individuals. Thus, a small increase in dose is less likely to result in serious toxic effects than a drug with a low therapeutic index.

Incorrect options:

- A. Determining the maximal effect of the drug: Graded dose-response curves
 - B. Determining the elimination rate of the drug: Drug concentration-time curves
 - C. Determining the relative potency of the drug: Graded dose-response curves
-

11. A hypotensive patient in the ICU is given an intravenous infusion of an agonist called phenylephrine, which mimics the actions of the natural ligand, epinephrine. The drug acts on receptors in the blood vessels, causing vasoconstriction, which increases blood pressure. The drug is a small molecule chemically similar to the natural ligand and produces the maximal effect.

Which of the following properties ultimately determines phenylephrine's pharmacologic effect (the change in biologic action)?

- A. Hydrophobicity
- B. Potency
- C. Selectivity for vascular receptors
- D. Structure-activity relationship*

(Select the one BEST answer.)

Rationale: D. Structure-activity relationship

The interaction of a drug with its therapeutic target is related to the drug's structural properties.

It is the chemical structure that determines all of the other characteristics of the drug – pharmacokinetics – absorption, distribution, metabolism, and excretion (ADME) – and pharmacodynamics properties – affinity, intrinsic activity, and efficacy.

Thinking it through:

- The chemical structure determines the physicochemical properties of a drug molecule.

A drug must have the size, shape, steric and electrical features necessary to ensure optimal noncovalent interactions necessary for drugs to exert their effects.

In case you are wondering about it, here is how it works (no need to memorize):

A drug and a specific biologic target assemble to form the larger drug-receptor complex through noncovalent interactions, such as hydrogen bonds, van der Waals forces, and electrostatic interactions.

These are called supramolecular interactions, which determine the structure and function of the overall assembly of the molecules.

The supramolecular interactions are essential for drug-receptor recognition and binding and determine the selectivity of the drug for the intended receptor, the specific pharmacologic effects, and the reversibility of the binding.

Incorrect answers: How can we eliminate these options to come up with the correct answer?

Options A, B, and C are not incorrect – but they are not the one BEST answer.

The molecular structure determines the drug's physicochemical properties, such as hydrophobicity and its receptor selectivity.

Potency refers to the dose of drug needed to produce a specific effect relative to a comparator. Potency is related to affinity, which is determined by the structure-activity relationship of the drug.

Structure-activity relationship is the BEST answer.

12. A 48-year-old man presents to the family practice clinic for pharmacologic management of hypertension and hypercholesterolemia, which have not adequately responded to diet and exercise. The patient has a family history of cardiovascular disease – his father died of a heart attack at age 63 and his mother died of a stroke at age 72. In addition to blood pressure medication, the patient agrees to therapy with a drug that reversibly binds to the active (catalytic) site of a specific enzyme and blocks the hepatic synthesis of cholesterol.

i) What is the mechanism of action of the prescribed drug?

ii) Draw the graph that describes the mechanism of the action experimentally and label it.

Hint #1: Notes page 12.

Hint #2: Highlight the clues that relate to pharmacodynamics concepts. Please don't become distracted by the other information.

Rationale: Competitive antagonist

Clues in the stem of the case vignette:

Reversibly binds

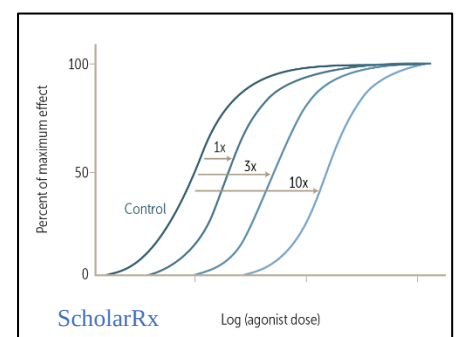
Active site

Blocks synthesis = blocks substrate

Competitive Antagonist

Thinking it through:

- The drug binds to the active site, which is the catalytic site of the enzyme. Thus, the drug has affinity.
- The drug binding blocks the enzyme's natural ligand (substrate) from occupying the site and does not activate the receptor (synthesis of cholesterol is inhibited). The drug has zero intrinsic activity. Thus, the drug is an antagonist.



- The binding is reversible. Therefore, the natural ligand (or other agonist) can access the site, given a high enough concentration, and produce its full effect.
- Thus, the drug is a competitive antagonist.

The characteristic pattern of competitive antagonism on a graded dose-response curve is a shift to the right with an increased concentration of agonist, which outcompetes the antagonist at the receptor, followed by an increased concentration of antagonist that inhibits the agonist – and so on.

The agonist achieves the maximal response when an adequate concentration is present at the receptor sites to outcompete the antagonist.

The antagonist reduces the potency of the agonist because a higher concentration of the agonist (higher EC₅₀) is needed to produce the effect.

Figure: ScholarRx Brick: General Pharmacology . Receptor Agonists and Antagonists

<https://exchange.scholarrx.com/brick/receptor-agonists-and-antagonists>

13. A 23-year-old medical student spending the summer in a malaria-endemic region takes an antimalarial drug for the prophylaxis (prevention) of malaria. The drug is 99% bound to plasma proteins. What is the change in function that can be expected from this drug-protein interaction? Short answer.

Rationale: No change. Plasma proteins are inert.

Many drugs have affinity for plasma proteins, which have no intrinsic regulatory function; they are inert.

❖ **Practice point:** Drug binding to inert proteins and macromolecules affects distribution of drug to the target. *Only unbound (free) drug is available to cross physiologic barriers and reach its target of action where it produces its pharmacological effect.*

If there is a substantial change in the concentration of free drug due to disease, the concentration of drug at the receptor site can change, altering response.

For example, a decrease in plasma proteins due to hepatic disease can result in an increase in free fraction of drug → increased concentration of drug at the target → increase magnitude of effect and risk of toxicity.

pharmacokinetics ↔ pharmacodynamics

14. A 34-year-old female with generalized anxiety disorder is prescribed Xanax (alprazolam). The drug acts by binding an allosteric site (BZ receptor) on GABA-A receptor, which is an ion channel. The drug enhances the receptor response to the natural ligand, GABA, which induces antianxiety and sedative effects.

- What is the mechanism of action of the prescribed drug?
- Draw the graph that describes the mechanism of the action experimentally and label it.

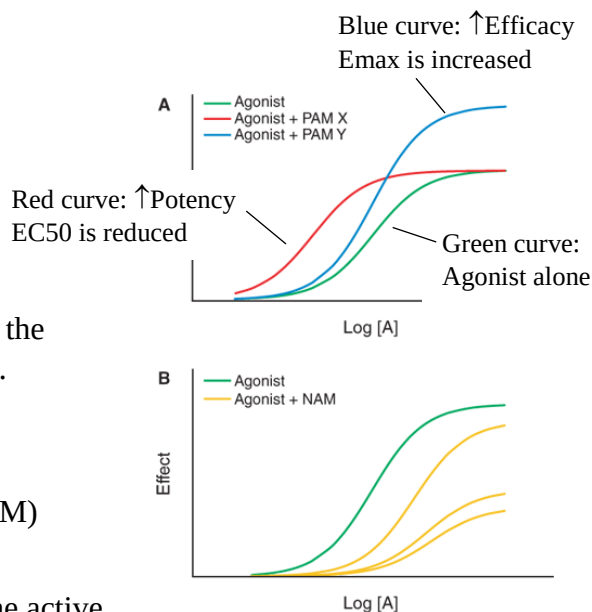
Rationale: positive allosteric modulator (PAM) on panel A

A positive allosteric modulator binds to an allosteric site and *enhances the affinity* of the receptor for the agonist, increasing the agonist's potency and/or effect.

Panel A: Positive Allosteric Modulator

Positive allosteric modulator X, PAM X, shifts the EC_{50} for the agonist leftward, which increases the potency of the agonist.

PAM Y increases the E_{max} of the agonist.



Source: Laurence L. Brunton, Björn C. Knollmann: Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e: Copyright © McGraw Hill. All rights reserved.

Compare with Panel B: Negative Allosteric Modulator (NAM)

Same concept but opposite effect.

NAM is a drug that binds a site on the receptor other than the active site and reduces the maximal effect of the agonist. A NAM causes a conformational change that reduces the affinity and/or the efficacy of the agonist, but may not entirely eliminate the response, depending on the degree of allosteric modulation.

In the set of dose-response curves in Panel B, the negative allosteric modulator, NAM, has a negative impact on both EC_{50} and E_{max} , reducing potency and efficacy.

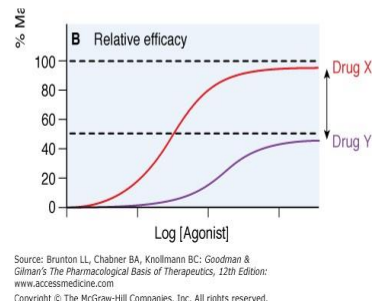
15. A study on isolated, perfused animal hearts showed that pindolol caused an increased heart rate, but to a lesser degree than that caused by the natural ligand. However, in the presence of a highly effective stimulant (agonist) acting on the same receptors, pindolol caused a dose-dependent reversible decrease in heart rate.

- What is pindolol's mechanism of action (classical receptor occupancy model)?
- Draw the graded dose-response graph that describes the mechanism of the action experimentally and label it.

Rationale: partial agonist / Drug Y on the chart below

The partial agonist is Drug Y in Panel B in the figure (classic receptor model).

- A partial agonist produces only a lesser response (lower effect) as compared to the full agonist (E_{max}). Higher doses do not achieve the maximal effect.
- In the presence of a full agonist (drug or natural ligand), a partial agonist behaves like a competitive antagonist because it prevents the full agonist from binding to the receptor and exerting the maximal response the receptor is capable of.



Partial agonists are sometimes called a partial agonist/antagonist because they activate the receptor but to a lower degree than the full agonist and, at the same time, block the binding of the full agonist.

16. In the two-state receptor occupancy model:

i) Which one of the following options would best describe pindolol's action?

- A. $R_a + D_a$
- B. $R_a + D_{pa}^*$
- C. $R_a + D_{ant} + R_i + D_{ant}$
- D. $R_i + D_i$

ii) Draw the graph that describes each of the options and label it.

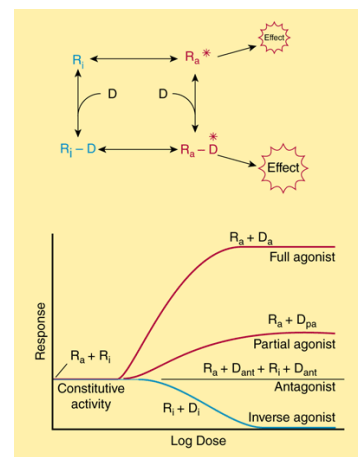
Rationale: Option B represents a partial agonist ($R_a + D_{pa}$) in the two state receptor model.

The two-state receptor occupancy model postulates that a receptor assumes two conformational states— active and inactive – in the absence of a ligand, and that a fraction of receptors in the receptor pool are in the inactive, nonfunctional form, R_i , and a fraction are in the activated form, R_a .

The extent to which equilibrium is shifted toward the active state is determined by the relative affinity of the drug for each of the conformations.

According to this model, a partial agonist binds to active R_a but also binds to the receptor in its inactive conformation, R_i , with lower affinity. The drug will produce a partial response, no matter how high the concentration of drug.

- Conventional full agonist drugs have a much higher affinity for the R_a conformation, and mass action favors the formation of the R_a -D complex with a much larger observed effect.



Katzung Figure 1-3

- Partial agonists have an intermediate affinity for both R_a and R_i forms.
- Neutral antagonists have equal affinity for both receptor forms, fixing R_a and R_i in the same relative amounts as in the absence of drug, so the drug will appear to be without effect, i.e. maintain the same level of constitutive activity.
- Inverse agonist has a much higher affinity for the R_i form and reduces constitutive activity, producing the opposite effect of the conventional agonist.

Comparing classical competitive antagonist and two-state model inverse agonist:

(Golan, et.al., 2017; Chapter 2)

- Both types of drugs reduce the activity of a receptor.
- In the presence of a full agonist, both competitive antagonists and inverse agonists reduce agonist potency.
- Competitive antagonists have no effect (zero intrinsic activity), whereas an inverse agonist deactivates receptors that are constitutively active in the absence of an agonist.
- Competitive antagonists prevent full, partial, and inverse agonists from binding the receptor.

17. A 79-year-old man is hospitalized for the treatment of decompensated heart failure. Continuous intravenous infusion of an inotrope is administered. The drug activates beta adrenergic receptors (G_s PCRs) in the heart, which increases strength of contraction, which leads to increased stroke volume and cardiac output. An effect of this drug is a reduction in response with continuous use of the drug due to endocytosis of the beta receptors.

What is the term used for this response to the continuous use of the inotrope in this scenario?

Short answer.

Hint: Notes p16

Rationale: Tachyphylaxis, tolerance, receptor desensitization, receptor downregulation are all correct answers.

Tachyphylaxis is defined as a rapidly diminishing response to successive doses of a drug, which reduces its effect.

Tolerance is defined as diminished response to repeated administration of a drug as the body adapts to the presence of the drug.

Cellular regulation of drug-receptor interactions by downregulation (internalization) of receptors prevents overstimulation that can harm the cell or the entire organism.

As a result of this physiologic regulation, many drugs alter the response to the agonist over time.

Desensitization is a phenomenon where receptors lose their responsiveness overtime. The mechanism is not clear. The effect may be due to phosphorylation of the receptor, endocytosis of the receptor-ligand complex, and/or downregulation.

18. The mother of an unconscious 4-year-old boy with a one-year-history of type 1 diabetes mellitus treated with insulin administers a subcutaneous injection of glucagon to reverse the hypoglycemia caused by an overdose of insulin.

Which of the following best describes the action of glucagon in the context of the pharmacology?

- A. Full agonist
- B. Partial agonist
- C. Competitive antagonist
- D. Noncompetitive antagonist
- E. Physiological antagonist*
- F. Chemical antagonist

Definitions:

Insulin is a hormone secreted by the endocrine pancreas; it stimulates cellular uptake of glucose, which reduces blood glucose concentrations.

Glucagon is a hormone secreted by the endocrine pancreas that stimulates glucose formation in the liver, which increases blood glucose.

Insulin and glucagon are both formulated for pharmacologic use.

Rationale: Glucagon is a physiological – functional – antagonist in this context.

Thinking it through:

The vignette describes hypoglycemic coma due to excessive insulin.

- The physiologic effect of *insulin* is to reduce blood glucose by stimulating cellular uptake.
- The physiologic effect of *glucagon* is to increase blood glucose by stimulating hepatic synthesis of glucose.
- The hypoglycemic effect of excess insulin is counterbalanced by the blood glucose increasing effects of glucagon.
- In other words, glucagon has the opposite **physiologic effect** of insulin, and their mechanisms of action are completely different → nonreceptor antagonism.

➤ **Practice point:** Glucagon is a treatment for hypoglycemia caused by glucose-lowering therapy (including insulin) in patients with impaired consciousness and no IV access (IV dextrose rapidly reversed hypoglycemia).

19. A 15-year-old male is prescribed oral tetracycline by the dermatologist for the treatment of acne. He is cautioned not to take the medication together with an antacid, dairy products, or vitamins with minerals. He should separate taking the antibiotic and the other products by at least 2 hours before or after taking the acne medication.

What mechanism of antagonism is described?

Short answer.

Rationale: Chemical antagonism

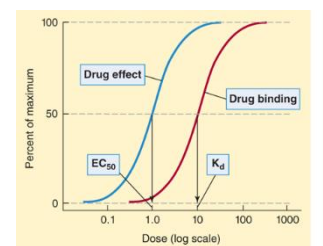
This drug interaction is caused by chemical antagonism that reduces absorption of the drug.

Multivalent cations chelate tetracycline (form a complex with the drug) preventing absorption from the gastrointestinal tract. Thus, less drug is delivered to the system circulation and drug effect is reduced. Indeed, the drug is likely to be ineffective.

20. An experimental agonist molecule tested in animal studies against an irreversible antagonist shows that its EC_{50} is $1/10^{\text{th}}$ (0.1%) of its K_d .

Which one of the following options best describes this finding?

- A. Binding to inert molecules
- B. Noncompetitive antagonism
- C. Partial agonism
- D. Spare receptors*



Source: Bertram G. Katzung, Marieke Knudsen-Hall, Anthony J. Trevor
Katzung & Trevor's Pharmacology: Examination & Board Review, Twelfth Edition
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Rationale: Spare receptors

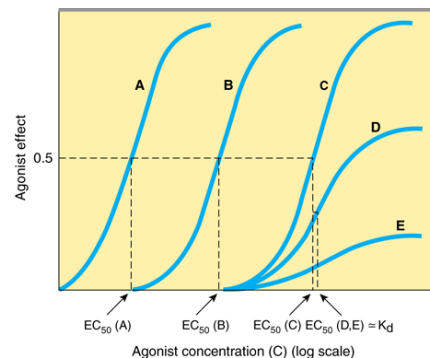
In circumstances where **only a fraction of the receptors in the receptor pool is needed to produce the maximal response** the remaining unbound receptors are referred to as spare receptors.

Spare receptors are illustrated experimentally using different concentrations of an irreversible antagonist.

Katzung Figure 2-2 shows the agonist concentration on a log scale on the x axis and the agonist effect on the y axis.

Curve A: Agonist response at concentration (A) in the absence of the antagonist.

Curve B: After treatment with a low concentration of the irreversible antagonist, the curve is shifted to the right.



Source: Katzung BG, Masters SB, Trevor AJ: Basic & Clinical Pharmacology, 12th edition: www.accessmedicine.com

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However, the agonist at concentration (B) achieves maximal responsiveness because the remaining available receptors still exceed the number required to produce the effect.

Katzung Figure 2-2

That is, even with the higher concentration of agonist, activation of only a fraction of receptors in the receptor pool is adequate to achieve $E_{\max}$: EC_{50} of agonist is still $< K_d$.

Curve C: After treatment with yet a larger concentration of antagonist, the agonist at concentration (C) achieves maximal response but the available receptors are no longer "spare". The number of remaining receptors is just enough to achieve the maximal response:

The agonist's $EC_{50} \approx K_d$.

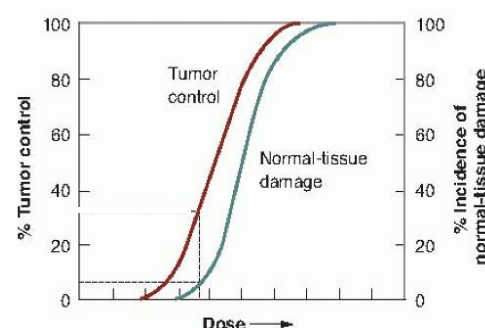
Curves D and E: Still higher concentrations of the irreversible antagonist reduce the number of available receptors to the point that response is diminished: $EC_{50} \approx K_d$. (Remember: K_d is the concentration at which half of the receptors are free and half are bound.)

Lower figure: Golan Figure 2-9 Chapter 2: Pharmacodynamics: Effect of a noncompetitive antagonist on the agonist dose-response curve in the presence of spare receptors.

21. Sixteen patients with epidermoid carcinomas of the head and neck received maximally tolerated doses of methotrexate alone as 30-hr intravenous infusions at 2-week intervals. 44% of patients responded. Significant bone marrow suppression and 4 deaths were incurred.

The cumulative distribution curve to the right shows that the drug has a low _____.

Fill in the answer.



Rationale: Therapeutic index, TI

The cumulative distribution curve shows that the drug has a low therapeutic index.

The ED_{50} of the dosage needed for tumor control is very near the LD_{50} , the dose that would cause death in 50% of the patients receiving the drug.

$TI = LD_{50} / ED_{50}$.

For the sake of illustration, estimating from the values on the graph, the ED_{50} might be 3.5 dosing units and the LD_{50} 5 dosing units. $TI = 5 / 3.5 = 1.4$, which is a very small TI.

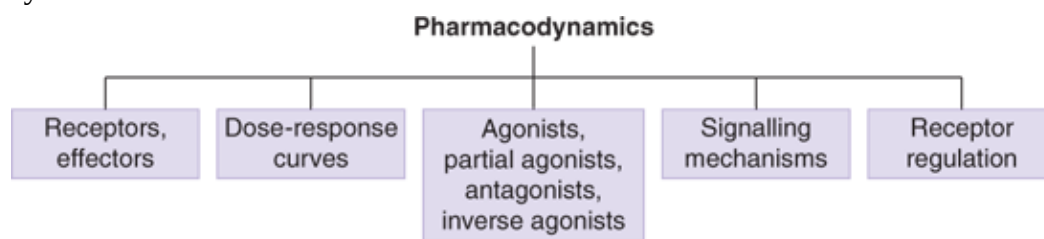
Cancer drugs are commonly associated with serious, even life-threatening toxicities. A small increase in dose increases the risk of the patient suffering the toxicities.

Source of case information: Levitt M, Moshe MB, et.al. Improved Therapeutic Index of Methotrexate with "Leucovorin Rescue". Cancer Research (1973) 33:1729-1734

<http://cancerres.aacrjournals.org/content/canres/33/7/1729.full.pdf>

SUMMARY

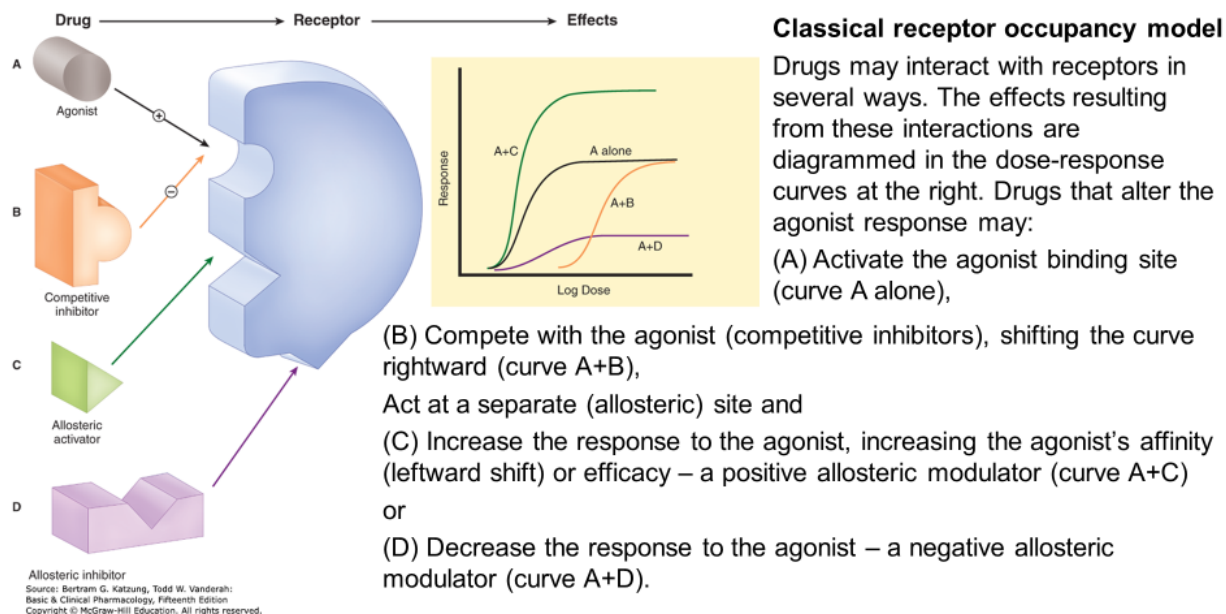
Pharmacodynamics: The branch of pharmacology that deals with the effects of drugs on biologic systems.



Source: A.J. Trevor, B.G. Katzung, M. Kruidering-Hall: Katzung & Trevor's Pharmacology: Examination & Board Review, 11th Ed. www.accesspharmacy.com
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- The physicochemical characteristics of a drug determine its ability to move through the body and initiate a cellular effect.
- The structure-activity relationship is the **relationship** between the chemical or 3D **structure** of a molecule and its biological **activity**.
- The interaction of a drug with its receptor is the fundamental event that initiates the action of the drug.
- The **extent** of drug action is determined by the location and functional capacity of receptors with which it interacts.
- The **duration** of drug action results from the duration of drug occupation of the receptor, intracellular signaling pathways, or receptor regulation.
- Drug binding to inert proteins/macromolecules can affect pharmacokinetic parameters of the drug but will not induce a change in the function of the biological system.
- Receptor occupancy model for explaining drug behavior assumes that response emanates from a receptor occupied by a drug.
- The drug effect is proportional to the number of receptors occupied and eliciting that effect, and the receptor occupancy is related to the ratio of the drug concentration and the equilibrium dissociation constant, K_d .
- A drug's affinity for a receptor is inversely proportional to its K_d .
- Graded dose-response curves depict the relationship between the concentration of a drug and the clinically observed response.
- The drug concentration and effect (indicated by the concentration of drug that produces a half-maximal effect, EC_{50}) and the drug concentration and receptor occupancy (indicated by the K_d) are related, as demonstrated by their respective graphic curves.
- Efficacy represents the *maximal effect* (E_{max}) of a drug at the highest tolerated dose, whereas potency reflects the *amount* of drug (the dose or concentration) required to cause a specific amount of effect, *eg*, the EC_{50} for a half-maximal effect. A drug may have high efficacy but low potency or vice versa.

- The response elicited by conformation-selective drugs is related to the drug's relative affinities for the active and/or inactive conformational states of the receptor.
- A maximal response elicited without agonist binding of the full complement of receptors describes the concept of spare receptors.
- Characteristic patterns of antagonists are associated with certain mechanisms of receptor blockade, i.e. competitive, irreversible and allosteric antagonism.



Citation: Chapter 1 Introduction: The Nature of Drugs & Drug Development & Regulation, Katzung BG, Vanderah TW. Basic & Clinical Pharmacology, 15e; 2021. Available at: <https://accessmedicine.mhmedical.com/content.aspx?bookid=2988§ionid=250580947> Accessed: June 22, 2021
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Slide adapted by LG

- Individuals vary in the magnitude of their response to the same dose (concentration) of a single drug or to similar drugs, and a given individual may not always respond in the same way to the same drug concentration.
- The therapeutic window is the range of steady state concentrations at which the likelihood of efficacy is high, and the probability of adverse effects is low.
- Monitoring plasma drug levels may be recommended when there is a(n):
 - relationship between plasma concentration and clinical effect
 - significant pharmacologic variability
 - established target concentration
 - narrow therapeutic window

Summary of Terms for Receptor Antagonists:

Competitive Antagonist

- **Binding Site:** Competes with the agonist for the same binding site on the receptor.
- **Reversibility:** Reversible; increasing the concentration of the agonist can overcome the antagonism.
- **Effect on Dose-Response Curve:** Shifts the dose-response curve to the right, meaning higher concentrations of the agonist are needed to achieve the same effect → reduces the potency ($\uparrow EC_{50}$) of the agonist.
- **Example:** Beta-blockers used to manage hypertension.

Irreversible and Pseudo-irreversible Active Site Antagonist

- **Binding Site:** Binds the same site as the agonist but forms a covalent bond, or very high-affinity bond, making the binding irreversible or functionally irreversible.
- **Reversibility:** Irreversible; the receptor is permanently inactivated until new receptors are synthesized.
- **Effect on Dose-Response Curve:** Reduces the maximum response ($\downarrow E_{max}$) that can be achieved by the agonist, as it decreases the number of available receptors.
- **Example:** Phenoxybenzamine, used in the treatment of pheochromocytoma.

Noncompetitive Allosteric Antagonist:

- **Binding site:** Binds a site near the agonist site and prevents activation of receptor signaling
- **Reversibility:** Reversible or irreversible
- **Effect on Dose-Response Curve:** Reduces agonist efficacy, E_{max} .

Negative Allosteric Modulator

- **Binding Site:** Binds to a different (allosteric) site on the receptor near the agonist's binding site.
- **Reversibility:** Can be reversible or irreversible, but typically reversible.
- **Effect on Dose-Response Curve:** Decreases the efficacy of the agonist by changing the receptor's conformation, making it less responsive to the agonist. This effect has a ceiling, meaning it can only inhibit the response to a certain extent.
- **Example:** Maraviroc, used as an HIV entry inhibitor.
Compare with Positive Allosteric Modulators potentiate the effect of the agonist ($\uparrow E_{max}$).

Uncompetitive Antagonist

- **Binding site:** Binds a site on an enzyme or receptor distinct from the substrate site; the site becomes accessible only after binding of the substrate to the active site
- **Reversibility:** Irreversible or pseudoirreversible
- **Effect on Dose-Response Curve:** Decreases both the potency and efficacy of the agonist; it produces a parallel decrease in response across concentrations
- **Example:** Ketamine binds the NMDA receptor only when activated by glutamate, which reduces Ca^{2+} influx, which reduces receptor signaling.

Resources:

Access Medicine Goodman & Gilman's The Pharmacological Basis of Therapeutics, 14e, 2024;
Chapter 3: Pharmacodynamics: Molecular Mechanisms of Drug Action

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 1: Introduction:
The Nature of Drugs & Drug Development & Regulation; Chapter 2: Drug Receptors &
Pharmacodynamics

LWW Health Libraries, Premium Basic Sciences, Lippincott Illustrated Reviews: Pharmacology,
8e, 2023; Chapter 2: Drug-Receptor Interactions and Pharmacodynamics

LWW Health Libraries, Premium Basic Science: Golan's Principles of Pharmacology: Chapter
2: The Pathophysiologic Basis of Drug Therapy, 4e, 2017

ScholarRx Brick: General Pharmacology > Receptor Agonists and Antagonists [https://usmle-
rx.scholarrx.com/rx-bricks/brick/CP_PHR0027](https://usmle-rx.scholarrx.com/rx-bricks/brick/CP_PHR0027)

Scholar Rx Bricks: Cellular and Molecular Biology> Cell-to-Cell Communications > Cell
signaling https://usmle-rx.scholarrx.com/rx-bricks/brick/CP_BCM0005